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ENGINEERING
BAYESIAN LEARNING AND MONTE CARLO SIMULATION

Final project:
Yacht Hydrodynamic Resistance

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1 Log–log Regression

1.1 Model and prior

The physics of slender sailing hulls suggests that, in the bulk regime, the residuary resistance grows as a power of the **Froude number**, resistance $\propto Fn^{\gamma_F}$ with γ_F close to 4. Taking logarithms turns this multiplicative law into a linear model, in which the exponent appears directly as a regression slope. We therefore fit

$$\log(\text{resistance}) = \beta_0 + \beta_F \log(\text{Froude}) + \sum_{k=1}^5 \beta_k \text{hull}_k + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where hull_k denotes the five geometric form factors. Since the smallest observed resistance is $0.01 > 0$, the logarithm is well defined for every record and no offset is required. Writing the $n = 308$ observations in matrix form as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\beta} \in \mathbb{R}^p$, $p = 7$, we adopt the non-informative **Jeffreys reference prior**

$$\pi(\boldsymbol{\beta}, \sigma^2) \propto \sigma^{-2}, \quad (2)$$

which lets the data dominate the inference and provides a transparent Bayesian baseline against which the model-selection and heteroscedasticity-corrected analyses of the following sections can be compared.

1.2 Posterior and Monte Carlo sampling

Under the prior (2) the posterior is Normal–Inverse–Gamma and is available in closed form. With $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ the least-squares estimate and $s^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 / (n - p)$ the unbiased error variance,

$$\sigma^2 \mid \mathbf{y} \sim \text{Inv-Gamma}\left(\frac{n-p}{2}, \frac{(n-p)s^2}{2}\right), \quad (3)$$

$$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y} \sim \mathcal{N}\left(\hat{\boldsymbol{\beta}}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}\right). \quad (4)$$

We sample the joint posterior by Monte Carlo: for each of $M = 20,000$ iterations we first draw σ^2 from (3) via a scaled inverse- χ^2 variate and then draw $\boldsymbol{\beta}$ from the Gaussian (4). The resulting sample $\{\boldsymbol{\beta}^{(m)}, \sigma^{2(m)}\}_{m=1}^M$ marginalises the regression coefficients over σ^2 (a multivariate Student- t) and is used to obtain all posterior means, standard deviations and 95% credible intervals, as well as the credible band of the fitted curve.

1.3 Results

Table 3 (in the appendix) reports the posterior summaries. The model explains the data very well ($R^2 = 0.97$, residual standard deviation $s = 0.32$ on the log scale). The posterior of the Froude exponent is sharply concentrated,

$$\beta_F \approx 4.69, \quad 95\% \text{ CI} = [4.60, 4.78],$$

confirming the expected power-law behaviour resistance $\sim Fn^4$ and verifying the project requirement $\hat{\beta}_F \approx 4$. Fig. 1 (left) shows the log–log scatter with the posterior-mean line and its 95% credible band: the band is extremely tight, reflecting the 300

observations and the strong signal of the Froude number. The Monte Carlo posterior of β_F is displayed in Fig. 1 (right); the whole posterior mass sits above the nominal value 4 ($\Pr(\beta_F > 4 \mid \mathbf{y}) \approx 1$), so the empirical exponent is best described as *slightly steeper* than the textbook value of four.

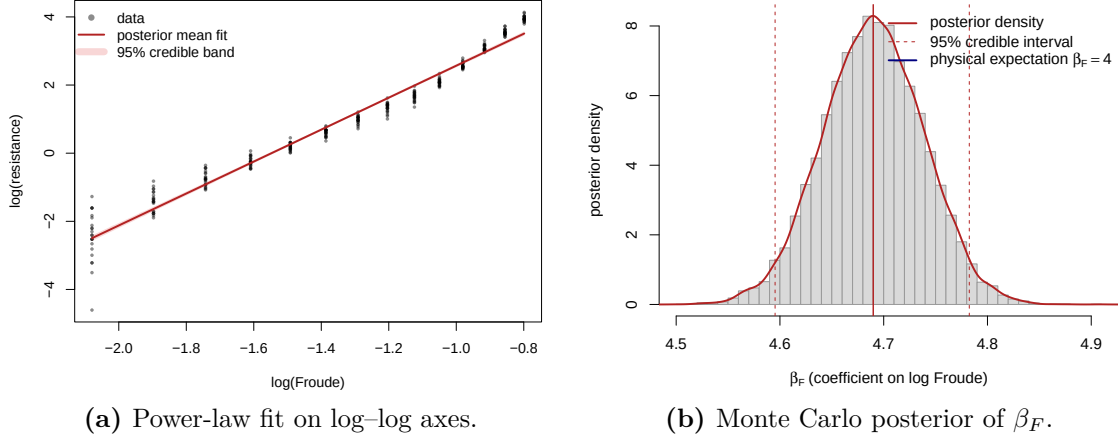


Figure 1: Left: observed $\log(\text{resistance})$ against $\log(\text{Froude})$ with the posterior-mean regression line (hull covariates held at their means) and its 95% credible band. Right: Monte Carlo posterior density of the Froude exponent β_F ; the dashed lines mark the 95% credible interval and the solid blue line the physical reference value $\beta_F = 4$.

Hull-geometry covariates. Once the Froude number is in the model, none of the five hull form factors is individually credible: every 95% interval comfortably contains zero (Table 3 and Fig. 3). The **prismatic** coefficient is especially poorly identified, with a posterior standard deviation of about 1.6, because the form factors vary only across the 22 hulls and are strongly collinear. This near-redundancy is precisely what the Bayesian model selection of the next section (BAS/BMA over the 2^5 submodels) is meant to resolve, by quantifying which covariates carry signal through their posterior inclusion probabilities rather than through single-model p -values.

Residual diagnostics. Although the global fit is excellent, the residuals reveal two systematic defects that the simple log–log model cannot capture (Fig. 2). First, the residuals-versus-fitted panel shows a clear U-shaped pattern: a single constant exponent slightly over-predicts at intermediate speeds and under-predicts at the highest speeds, consistent with the local Froude exponent increasing from ≈ 4.7 in the full range to above 6 in the high-speed subset. Second, the spread of the residuals is markedly larger at low fitted values, i.e. at low Froude numbers where the resistance is tiny and dominated by measurement noise — a textbook signature of heteroscedasticity, also visible as the fanning of the data cloud on the left of Fig. 1. The Normal Q–Q plot confirms heavier-than-Gaussian lower tails. These two findings (speed-dependent curvature and non-constant variance) directly motivate the heteroscedasticity-corrected nonlinear model developed in the subsequent section.

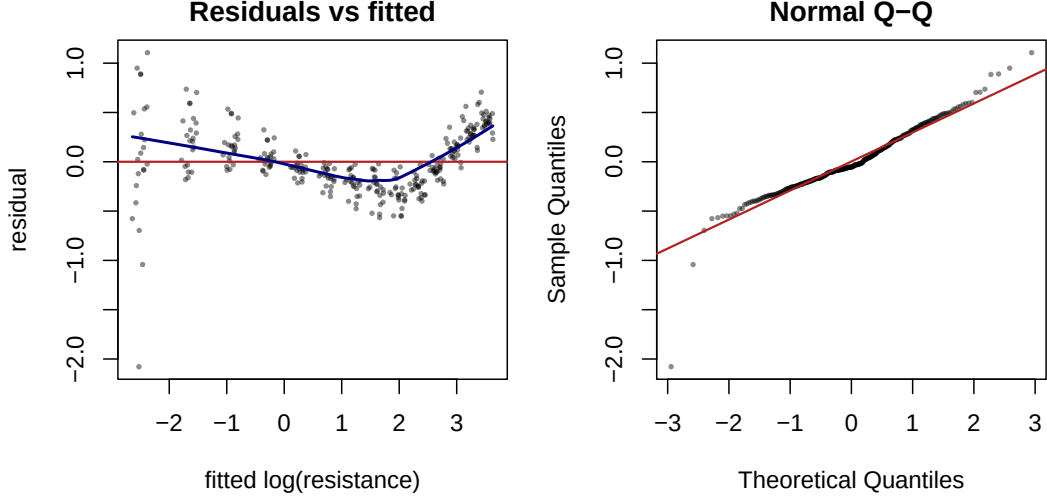


Figure 2: Residual diagnostics for the log-log model. Left: residuals versus fitted values, with a LOWESS trend (blue) revealing speed-dependent curvature and a variance that shrinks as the fitted value grows. Right: Normal Q-Q plot, showing heavier-than-Gaussian lower tails.

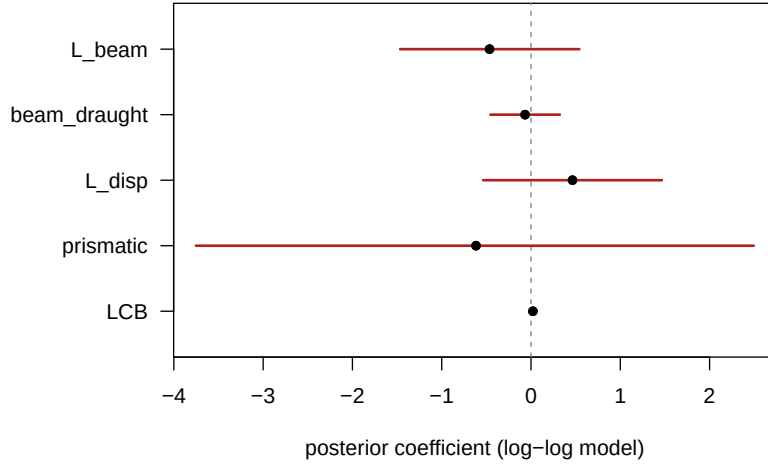


Figure 3: Posterior means and 95% credible intervals of the five hull-geometry coefficients in the log-log model. Every interval straddles zero (dashed line), indicating that no single form factor is resolved once the Froude number is accounted for.

1.4 Robustness to an informative prior

The reference prior (2) is deliberately objective: it lets the data alone estimate the Froude exponent, which we then *verify* against the physical value 4. A complementary, fully Bayesian alternative is to *encode* the physics directly through an informative prior on the exponent,

$$\beta_F \sim \mathcal{N}(4, \tau^2), \quad (5)$$

while keeping the remaining coefficients and σ^2 vague. The prior standard deviation τ controls how strongly the prior insists on the textbook value. Because the resulting

full conditionals remain in closed form — $\beta \mid \sigma^2, \mathbf{y}$ Gaussian and $\sigma^2 \mid \beta, \mathbf{y}$ Inverse-Gamma — the posterior is obtained with a short Gibbs sampler, and we sweep $\tau \in \{0.05, 0.1, 0.2, 0.5\}$.

Table 1 and Fig. 4 summarise the effect. With a diffuse prior the analysis reproduces the reference result ($\beta_F \approx 4.69$). As the prior is tightened around 4 the posterior mean is pulled downwards, but the data resist: only a near-dogmatic prior ($\tau = 0.05$, a prior 95% interval of roughly $[3.9, 4.1]$) drags the posterior mean as far as 4.33, and even then $\Pr(\beta_F > 4 \mid \mathbf{y}) = 1$. For any moderately diffuse physical prior ($\tau \geq 0.1$) the likelihood dominates and the posterior stays in the range 4.56–4.69. The conclusion that the empirical exponent is genuinely steeper than Fn^4 is therefore **robust to the prior**: it is a feature of the $n = 308$ observations rather than an artefact of the non-informative choice. Equivalently, this is a transparent instance of prior–data conflict, in which a sufficiently informative likelihood overwhelms all but an unreasonably confident prior.

Table 1: Posterior of the Froude exponent β_F under an informative prior $\mathcal{N}(4, \tau^2)$ of increasing strength, obtained by Gibbs sampling, compared with the reference-prior result.

Prior on β_F	Mean	2.5%	97.5%	$\Pr(\beta_F > 4)$
$\mathcal{N}(4, 0.05^2)$	4.33	4.26	4.41	1.00
$\mathcal{N}(4, 0.10^2)$	4.56	4.47	4.64	1.00
$\mathcal{N}(4, 0.20^2)$	4.65	4.56	4.74	1.00
$\mathcal{N}(4, 0.50^2)$	4.68	4.59	4.78	1.00
reference σ^{-2}	4.69	4.60	4.78	1.00

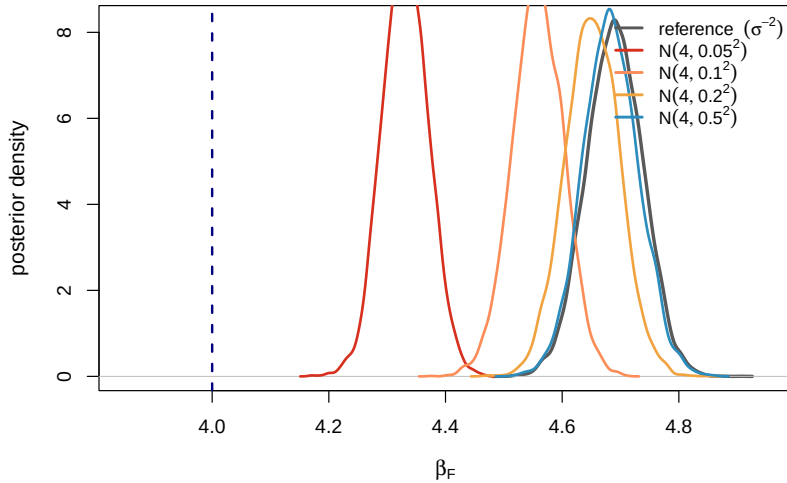


Figure 4: Posterior density of the Froude exponent β_F under informative priors $\mathcal{N}(4, \tau^2)$ of increasing strength (coloured) and under the reference prior (grey). The dashed blue line marks the physical value $\beta_F = 4$. Tightening the prior toward 4 shifts the posterior only modestly; the data keep it well above 4 unless the prior is unreasonably confident.

2 BAS

3 Heteroscedasticity-Corrected Nonlinear Model

A Tables

Table 2: Variables of the yacht hydrodynamics dataset.

Variable	Description
LCB	Longitudinal position of the centre of buoyancy
prismatic	Prismatic coefficient
L_disp	Length–displacement ratio
beam_draught	Beam–draught ratio
L_beam	Length–beam ratio
Froude	Froude number, $Fn = v/\sqrt{gL}$
resistance	Residuary resistance per unit weight of displacement

Table 3: Posterior summary of the log–log model (1) under the reference prior ($M = 20,000$ Monte Carlo draws). $\Pr(> 0)$ is the posterior probability that the coefficient is positive.

Coefficient	Mean	SD	2.5%	97.5%	$\Pr(> 0)$
Intercept β_0	7.183	0.977	5.245	9.122	1.00
log(Froude) β_F	4.690	0.048	4.595	4.782	1.00
LCB	0.021	0.012	−0.003	0.046	0.96
prismatic	−0.616	1.597	−3.753	2.495	0.35
L_disp	0.464	0.512	−0.536	1.466	0.82
beam_draught	−0.067	0.200	−0.455	0.324	0.37
L_beam	−0.464	0.515	−1.466	0.542	0.18